# A Root-Marked Three-to-Six Theorem for Approachable Infinite Omnibus Walks

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**Abstract**

We give a Lean 4 proof of a precisely delimited version of a three-to-six sonority bound inspired by Pyle's Conjecture 1. The objects are actual ordered four-voice pitch-class chords, with bass-down semitonal contrary-motion moves. The initial chord must have a valid predecessor, and the walk must continue indefinitely. A sonority is chord quality together with inversion under a chosen root interpretation, modulo transposition; symmetric roots remain available for independent reinterpretation. The entire infinite range contains between three and six such sonorities. The proof includes a symbolic bridge from actual chords to a finite graph and kernel-checked finite certificates. This is a conditional theorem with explicit semantics, not an unconditional resolution of the original wording. Established mathematical novelty is not claimed.

## 1 Scope and relation to the source

Pyle [1] defines four-voice triads, seventh chords and three descending-bass omnibus moves, and proposes a bound of three to six distinct sonorities before an extension. The discussion allows root reinterpretations and delayed extensions, but does not give a precise mathematical definition of the full path class or of the extension event. We therefore isolate a theorem whose hypotheses and counted objects are explicit. The interpretive question retained for later correspondence appears in Appendix A.

The present result counts *quality and inversion*, not absolute pitch-class roots, literal voicings, chord occurrences, or the length of a repeated block. It is restricted to Definition 6's fixed bass-down direction. Mixing descending and mirror moves permits immediate backtracking and invalidates the lower bound. No mixed-direction result is asserted.

## 2 Actual chords and root-marked readings

Let a chord be an ordered tuple $c = (b, t, a, s) \in (\mathbb{Z}/12\mathbb{Z})^4$, with coordinates corresponding to bass, tenor, alto and soprano. Let $\mathcal{Q}$ consist of those tuples for which some reordering has one zero successive interval and two intervals in $\{3, 4\}$, or has four distinct entries and three successive intervals in $\{3, 4\}$. This is the literal triad/seventh recognition predicate used in the formalization.

For $v \in \{T, A, S\}$, define $f_v$ by decreasing the bass by one, increasing upper voice $v$ by one, and fixing the other two voices, all modulo twelve. An admissible step $c \longrightarrow d$ means $c, d \in \mathcal{Q}$ and $d = f_v(c)$ for some $v$. A chord is *approachable* if it has an admissible predecessor.

For each quality $q$, let $D_q$ be its ordered third-stack template:

| $q$ | $M$ | $m$ | dim | $+$ | $M7$ | $7$ |
|---|---|---|---|---|---|---|
| $D_q$ | (047) | (037) | (036) | (048) | (047 11) | (047 10) |

| $q$ | $mM7$ | $m7$ | $\varnothing 7$ | $\circ 7$ | $+M7$ |
|---|---|---|---|---|---|
| $D_q$ | (037 11) | (037 10) | (036 10) | (0369) | (048 11) |

The compact notation lists pitch-class offsets; for example (047 11) means $(0, 4, 7, 11)$.

**Definition 1.** *A* root-marked harmonic type *is* $(q, i)$ *with* $0 \le i < |D_q|$. *It reads chord* $c$ *if*

$$\{0, t-b, a-b, s-b\} = \{d - D_q[i] : d \in D_q\} \quad \text{in } \mathbb{Z}/12\mathbb{Z}.$$

*Thus the chosen root is* $b - D_q[i]$, *while only* $(q, i)$ *is counted.*

This definition ignores upper-voice order and doubled-note choice. It does not identify the three inversion readings of an augmented triad or the four inversion readings of a fully diminished seventh. Each occurrence can choose any valid reading, independently of preceding occurrences. The underlying chord still has to satisfy the literal four-voice predicate, so set equality does not admit extra chord types or invalid multiplicities.

# 3 The conditional theorem

**Theorem 2** (Entire-range three-to-six bound)**.** *Let* $c : \mathbb{N} \to \mathcal{Q}$ *be an infinite admissible bass-down walk. Suppose that* $c(0)$ *is approachable. For each* $n$, *choose any root-marked harmonic type* $h(n)$ *reading* $c(n)$. *Then*

$$3 \ \le \big|\{h(n) : n \in \mathbb{N}\}\big| \ \le 6.$$

*In fact,* $h(0), h(1), h(2)$ *are pairwise distinct.*

This bounds the complete range, including types introduced after any selected extension boundary. It consequently bounds every prefix from above; the three-type lower bound applies once three consecutive positions are counted. No particular musical definition of an extension is used in its proof.

## 3.1 Reduction to intrinsic finite states

Normalize a chord by subtracting its bass and sorting the three upper coordinates. Write

$$\sigma(c) = \mathrm{sort}(t-b, a-b, s-b).$$

The intrinsic state set is $\mathcal{S} = \{(x, y, z) : 0 \le x \le y \le z < 12, (0, x, y, z) \in \mathcal{Q}\}$. It has 55 elements. Successors are computed by applying all three literal moves to $(0, x, y, z)$, retaining admissible results, and normalizing again.

The reused symbolic bridge proves, for every actual valid chord $c$,

$$q \in \mathrm{succ}(\sigma(c)) \quad \Longleftrightarrow \quad \exists d \ (c \longrightarrow d \ \wedge \ \sigma(d) = q).$$

Translation invariance and upper-voice permutations establish this equality; it is not assumed from a graph drawing. Node completeness is separately checked in twelve finite slices. The direct unsorted voice-relative reading predicate is proved equivalent to the sorted-state predicate.

### 3.2 Certificates and proof

Let $K$ be the union of the following four state lists:

$$C_1 = \{(0,4,7),(1,5,9),(2,5,8),(3,6,10),(3,7,9),(4,8,11)\},$$
$$C_2 = \{(0,4,8),(2,5,9),(3,7,10)\},$$
$$C_3 = \{(0,5,8),(1,4,8),(2,6,9),(3,5,9),(3,7,11),(4,7,10)\},$$
$$C_4 = \{(1,5,8),(3,6,9),(4,7,11)\}.$$

These lists are certificate witnesses, not alternative definitions of admissible chords or moves. Three finite propositions are evaluated by Lean's kernel from the intrinsic definitions:

1. If $p \in \mathcal{S}$ and $p \to q \to r \to s$, then $q \in K$.
2. For each $C_j$, a successor of a state in $C_j$ that belongs to $K$ also belongs to $C_j$. The union of all valid marked readings of states in $C_j$ has at most six elements.
3. For $p \to q \to r$ with $p, q, r \in K$, every choice of marked readings at the three positions gives three pairwise distinct types.

*Proof of Theorem 2.* Project the actual walk to $\sigma(c(n))$. At its first position, use the assumed predecessor and the next two moves in certificate 1. At any later position, use the preceding position and the next two moves. Hence every projected state belongs to $K$.

Choose a component list $C_j$ containing the initial state. Certificate 2 and induction keep the entire projected walk in that same list. Every marked reading then lies in its computed union of at most six types. Certificate 3 applied to the first two edges gives three distinct types. The direct-reading bridge transfers both conclusions to the original ordered chord walk. □

The component capacities, with all symmetric-root choices included, are:

| List | Marked quality/inversion inventory | Capacity |
|---|---|---|
| $C_1$ | $M_0,\ (+M7)_0, (+M7)_3,\ (\varnothing 7)_0, (\varnothing 7)_1, (\varnothing 7)_3$ | 6 |
| $C_2$ | $+_0, +_1, +_2,\ (m7)_0, (m7)_3$ | 5 |
| $C_3$ | $7_0, 7_2, 7_3,\ m_2,\ (mM7)_0, (mM7)_3$ | 6 |
| $C_4$ | $(\circ 7)_0, (\circ 7)_1, (\circ 7)_2, (\circ 7)_3,\ (M7)_0, (M7)_3$ | 6 |

Subscripts are inversion indices, with zero denoting root position.

## 4 Why the hypotheses are material

Consider the actual ordered chords

$$(0,0,3,6),\ (11,0,3,7),\ (10,1,3,7),\ (9,1,4,7),$$
$$(8,1,4,8),\ (7,1,4,9),\ (6,1,5,9).$$

Their moves are $S, T, A, S, S, A$. Their quality/inversion types are, in order,

$$\dim_0,\ (mM7)_3,\ 7_2,\ 7_0,\ m_2,\ 7_3,\ (mM7)_0.$$

All seven are distinct and have unambiguous roots. The final state admits an infinite admissible continuation; the first chord has no admissible predecessor. Thus simply discarding the initial-approachability hypothesis would make the upper bound false even for uniquely rooted chords.

The optional formal witness supplement verifies a 36-chord actual periodic walk attaining six types, and another attaining exactly three. It also verifies the seven-type example above with an infinite continuation and proves that its initial chord has no valid predecessor. Thus both endpoints of the conditional bound are attainable.

A finite approachable walk can instead leave its recurrent family and end at an additional type. Therefore approachability alone is insufficient if indefinite continuation is dropped. These observations explain why our hypotheses are stated visibly rather than inferred from the word "progression." They are not by themselves counterexamples to a uniquely specified interpretation of Pyle's conjecture.

## 5 Formal implementation and verification boundary

The core proof uses Lean 4.32.2 and its standard library. It freshly reuses 23 source modules from the earlier public project [2], ending in `NodeJoin` and `LocalQuotientLift`; no earlier compiled project objects are needed. The new modules provide the main result and concrete witnesses:

- `RootMarkedBounds`: template readings, finite certificates, and the witness-plus-list-cover theorem `three_to_six`.
- `RawReading`: direct actual-voice reading equivalence.
  The export is `raw_three_to_six`.
- `Witnesses`: literal periodic chord arrays, universal periodic path proofs, endpoint attainment, and the unapproachable seven-type example.
- `Cardinality`: a Mathlib wrapper producing a finite set equal to the whole infinite range, with cardinality between three and six.

The core conclusion gives three distinct attained types and a list of length at most six covering every index; this is already the desired cardinality bound. The wrapper merely expresses it with `Finset.card`.

The printed axioms of the three main bound exports are precisely

`propext`, `Classical.choice`, `Quot.sound`.

There are no proof placeholders, custom axioms, or native-evaluation axioms in these exports. Finite checks use `decide`. The supplied rebuild script separates the standard-library proof from the optional Mathlib cardinality presentation. Build success establishes the exact formal statement above; it does not establish a priority claim or identify an unstated musical definition with a formal predicate.

## A Question reserved for later correspondence

Does the seven-sonority admissible prefix in Section 4 count as a generalized omnibus progression before its first extension? If not, which chord or transition excludes it, or constitutes the extension?

The point is to determine whether the author's intended class excludes the unapproachable initial chord, imposes recurrence or another condition, or uses a different extension boundary. The theorem in this note remains valid regardless of the answer, while its precise relationship to the original conjecture depends on that clarification. No email or publication is part of this deliverable.

## References


[1] Hayden Pyle. Generalized Omnibus Progressions: A Mathematical Formalization of Harmonic Progressions Exhibiting Semitonal Contrary Motion and Tertian Sonorities. *Journal MusMat* IX (2025), 27–41. DOI: 10.46926/musmat.2025v9.27-41. https://musmat.org/journal/v9/02-Pyle.pdf.

[2] GPT-6 Astra Musical Edition project. Omnibus formalization, public snapshot 80a1a270d9355783c6c856bc463ebc936c5ae5c5. https://github.com/NoahWLono/GPT-6-Astra-Musical-Edition. The reused modules provide the literal chord/move model and symbolic quotient bridge. Their earlier graph and image-count results are not presented here as new discoveries.